I. Hosting of Application Components

1- Host Site

The application components will be deployed using a cloud-based virtual machine environment.

- Target Cloud Platform: Amazon Web Services (AWS EC2)
- Operating System: Ubuntu Linux Server
- Deployment Model: Microservices-based deployment
- Each service will run as an independent process on separate ports:
    1. Log Ingestion Service → Port 8000
    2. Log Processing Service → Port 8001
    3. Detection Service → Port 8002
    4. Incident Management Service → Port 8003
    5. Response Automation Engine → Port 8004

During development, services are hosted locally. For production, they will be deployed on a cloud VM instance.

2- Deployment Strategy

The following deployment strategy will be adopted:

Server Configuration

- Launch Ubuntu-based EC2 instance.
- Install Python 3 and required dependencies.
- Create isolated virtual environments for each microservice.
- Configure firewall rules to allow only required ports (e.g., 80, 443, 8000–8004).

Application Deployment Steps

1. Upload source code to the server using Git.
2. Create virtual environment for each service.
3. Install dependencies using pip install -r requirements.txt.
4. Start services using uvicorn with designated ports.
5. Configure internal API URLs:
    - Ingestion Service forwards logs to Processing Service.
    - Processing Service sends structured logs to Detection Service.
    - Detection Service communicates with Incident Management.
6. Use a reverse proxy (e.g., Nginx) to route external traffic.

7. Optionally use Docker containers for containerized deployment.

This ensures independent deployment and scalability of services.

3- Security Measures (Optional)

Since the system processes log data, the following security mechanisms will be applied:

- HTTPS encryption using SSL/TLS for secure communication.
- Firewall rules to restrict access to only required ports.
- API validation to prevent malformed or malicious inputs.
- Role-based access control for administrative endpoints.
- Logging and monitoring for intrusion detection.

These measures ensure secure communication and protect system integrity.